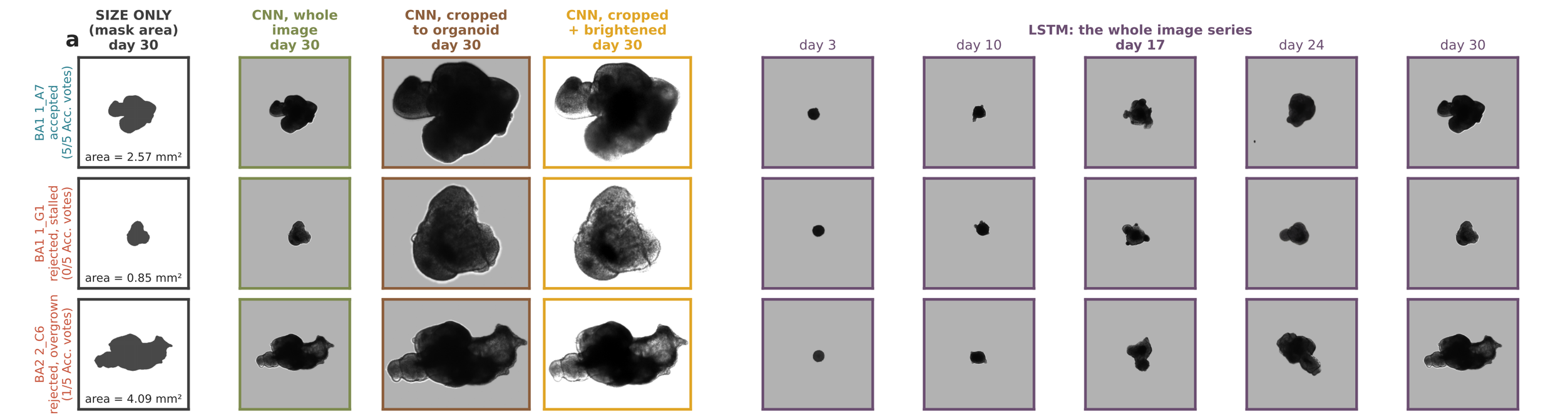
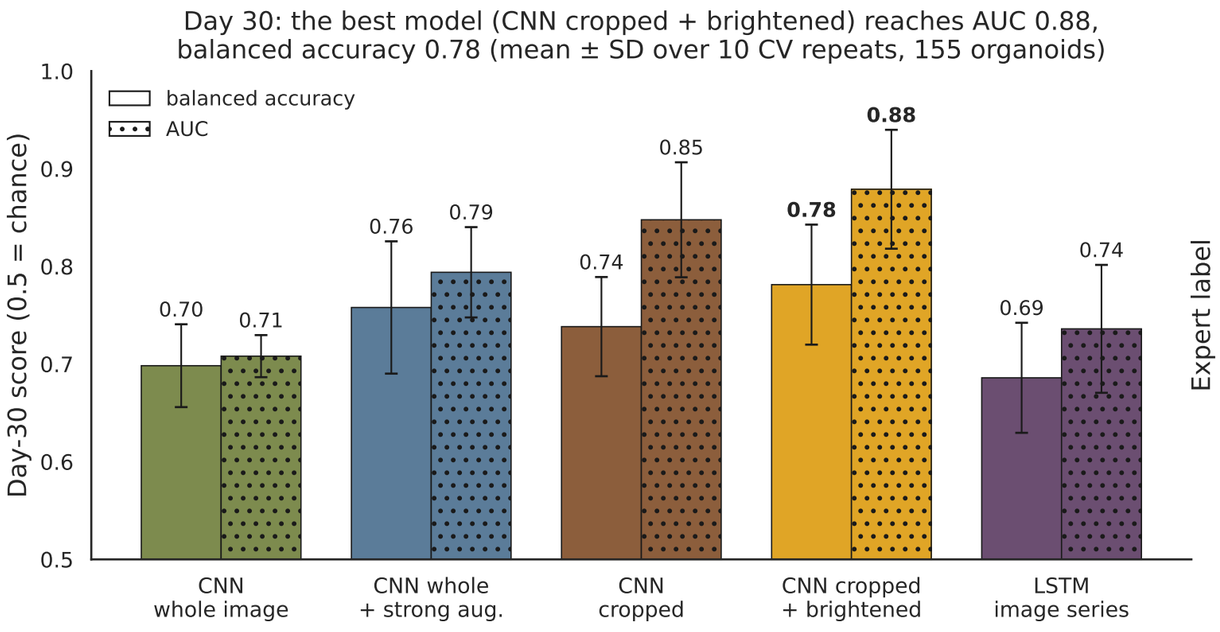
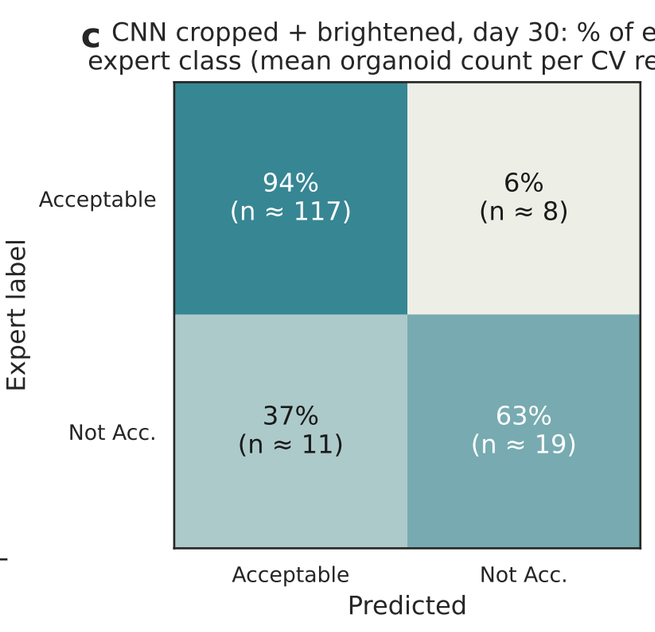
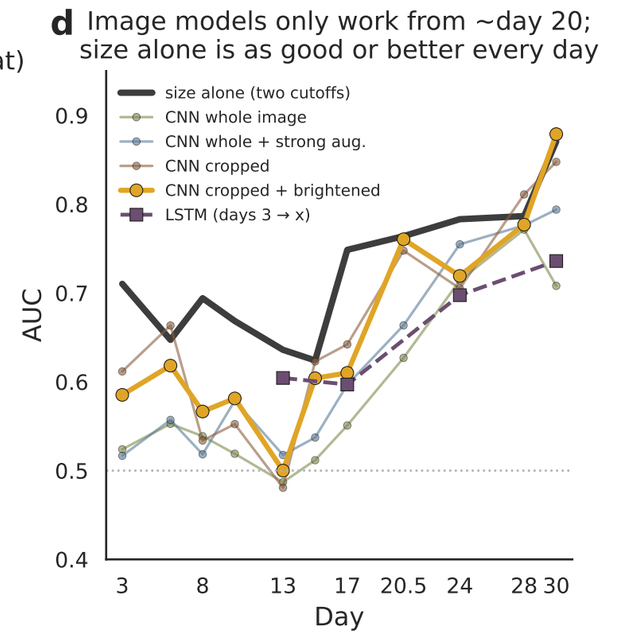
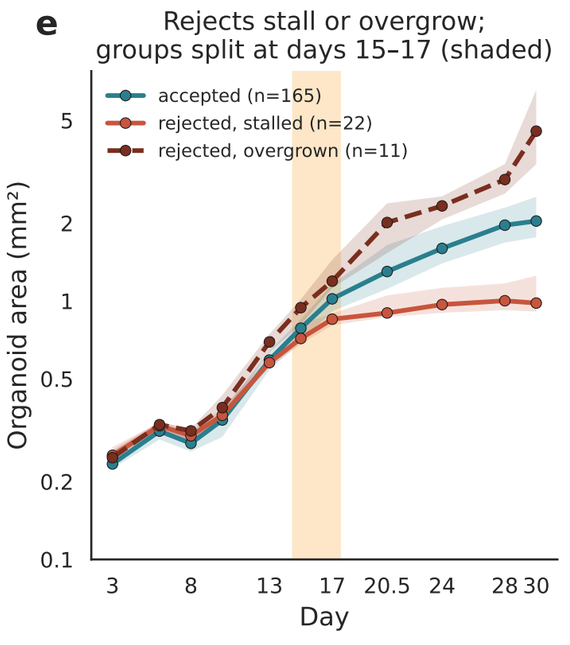
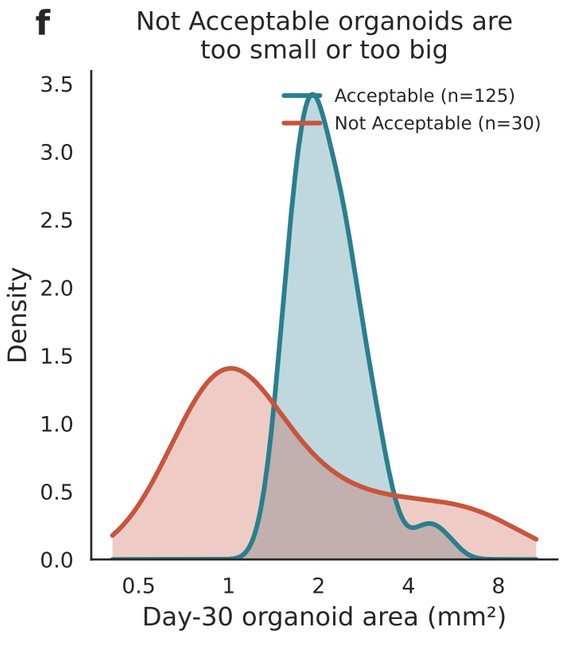

a
SIZE ONLY (mask area) day 30
CNN, whole image day 30
CNN, cropped to organoid day 30
CNN, cropped + brightened day 30
LSTM: the whole image series
day 3
day 10
day 17
day 24
day 30
BA1 1_A7 accepted (5/5 Acc. votes)
area = 2.57 mm²
BA1 1_G1 rejected, stalled (0/5 Acc. votes)
area = 0.85 mm²
BA2 2_C6 rejected, overgrown (1/5 Acc. votes)
area = 4.09 mm²

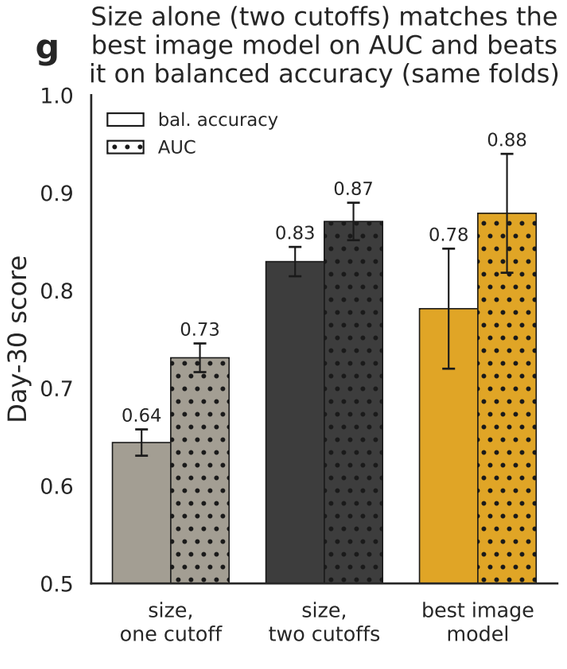

b
Day 30: the best model (CNN cropped + brightened) reaches AUC 0.88, balanced accuracy 0.78 (mean ± SD over 10 CV repeats, 155 organoids)
balanced accuracy
AUC
Day-30 score (0.5 = chance)
0.70
0.71
0.76
0.79
0.74
0.85
0.78
0.88
0.69
0.74
CNN whole image
CNN whole + strong aug.
CNN cropped
CNN cropped + brightened
LSTM image series

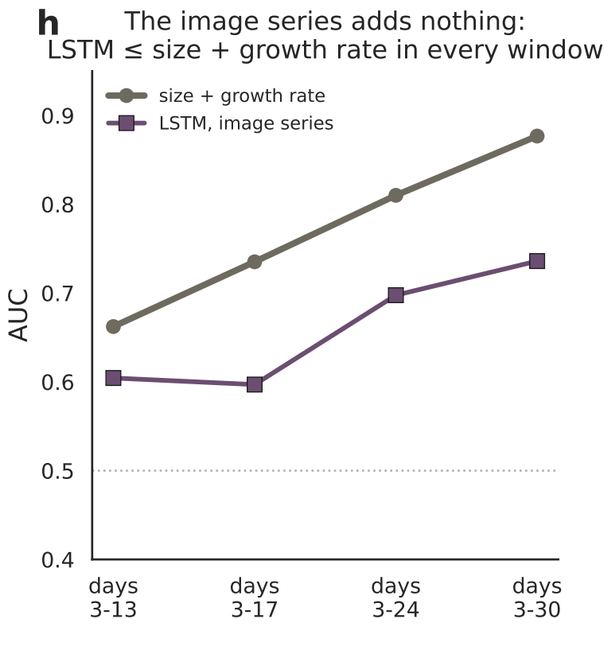

c
CNN cropped + brightened, day 30: % of each expert class (mean organoid count per CV repeat)
Acceptable
Not Acc.
94% (n ≈ 117)
6% (n ≈ 8)
37% (n ≈ 11)
63% (n ≈ 19)
Expert label
Acceptable
Not Acc.
Predicted


d
Image models only work from ~day 20; size alone is as good or better every day
size alone (two cutoffs)
CNN whole image
CNN whole + strong aug.
CNN cropped
CNN cropped + brightened
LSTM (days 3 → x)
AUC
Day


e
Rejects stall or overgrow; groups split at days 15–17 (shaded)
accepted (n=165)
rejected, stalled (n=22)
rejected, overgrown (n=11)
Organoid area (mm²)
Day


f
Not Acceptable organoids are too small or too big
Acceptable (n=125)
Not Acceptable (n=30)
Density
Day-30 organoid area (mm²)


g
Size alone (two cutoffs) matches the best image model on AUC and beats it on balanced accuracy (same folds)
bal. accuracy
AUC
Day-30 score
0.64
0.73
0.83
0.87
0.78
0.88
size, one cutoff
size, two cutoffs
best image model


h
The image series adds nothing: LSTM ≤ size + growth rate in every window
size + growth rate
LSTM, image series
AUC
days 3-13
days 3-17
days 3-24
days 3-30